



Variable Basics in JavaScript

Brief Overview

This note covers **JavaScript variable usage** and was created from a [5-page PDF](#). It explains **declaring** variables, reassigning values, using alert and prompt, and string concatenation with a simple game level example.

Key Points

- Declaring and initializing variables
- Reassigning values
- Interacting via alert and prompt
- Concatenating strings and a basic game counter



Declaring a Variable

A **variable** is a named container that stores a piece of data in memory.

- Syntax: `var variableName = value;`
- Example in JavaScript:

```
var myName = "Angela";
```

- After running the code, typing `myName` in the console returns "Angela".



How the Syntax Works

The structure of a variable declaration in JavaScript.

Part	Description
<code>var</code>	Keyword that creates a new container (similar to alert or prompt).
<code>myName</code>	Identifier – the name given to the container.

=	Assignment operator – places the value inside the container.
"Angela"	The value (a string) stored in the container.
;	Terminates the statement (acts like a lid on the box).

Changing a Variable

Reassigning a variable updates the data stored without using the var keyword again.

- To change the stored value:

```
myName = "Rebecca";
```

- The computer:
 - Finds the container named myName.
 - Removes the old content ("Angela").
 - Stores the new content ("Rebecca").
- Note:** var is required **only** when the variable is first created.

Challenge

Task: Show the content of myName using an alert **without typing any string literals**.

- Hint: Use the variable name directly inside the alert call.

Using Alerts

An alert displays a message in a dialog box.

- Basic usage:

```
alert(myName);
```

- Result: A dialog showing "Angela" (the current value of myName).

Using Prompt

A prompt asks the user for input and returns the entered string.

- Create a new variable to store the user's response:

```
var yourName = prompt("What is your name?");
```

- After the user types **Jack Bauer** and clicks **OK**, yourName holds "Jack Bauer".

+ Combining Variables

Concatenating variables with strings creates dynamic messages.

- Example:

```
alert("My name is " + myName + ", welcome to my course " + yourName + "!");
```

- Output: My name is Angela, welcome to my course Jack Bauer!

Practical Example

Using a variable to track game progress.

- Initialize a level counter:

```
var gameLevel = 1;
```

- When the player completes a level, update the counter:

```
gameLevel = 2;    // after level 1  
gameLevel = 3;    // after level 2
```

- Display the current level:

```
alert("Your level is currently: " + gameLevel);
```

- Result: Shows Your level is currently: 3.

Naming Rules (Preview)

Variables should have descriptive, camel-cased names (e.g., playerScore, totalPrice).

(Details to be covered in the next lesson.)